Series CC Overview and Cross-Paper Inheritance Framework

A derivation note from the CKS theory trilogy.

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Date: May 15, 2026

Attribution

This work derives from and formalizes commitments across the CKS theory trilogy: "The Instinct/Reasoning Separation Outside the Model" (Li, April 2026), "Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems" (Li, April 2026), and "Inter-Self Coordination via Shared Substrate / Full Aspect Integration" (Li, April 2026). It does not introduce new architectural commitments. Its contribution is to establish the framework for Series CC: the cross-paper inheritance notes that make explicit how Paper 3's inter-Self commitments derive from, and extend, the prior commitments of Papers 1 and 2.

Abstract

The CKS theory trilogy comprises three papers that address substrate-mediated coordination at progressively larger scopes: within a cell (Paper 1), within a Self (Paper 2), and across the inter-Self perimeter (Paper 3). A reader approaching only Paper 3 could plausibly treat its architectural commitments as a fresh set of claims requiring independent defense. They are not. Each Paper 3 commitment is a direct extension of a specific Paper 1 or Paper 2 commitment to the inter-Self scope—a scope extension, not a conceptual departure. Series CC is the set of derivation notes that establishes this inheritance chain explicitly. This note, CC.00, serves as the Series CC overview: it states the prior-art function of the inheritance chain, describes the two subseries (CC.1 covering Paper 3 ↔ Paper 1 inheritance, CC.2 covering Paper 3 ↔ Paper 2 inheritance), specifies the consistent five-element structure used in every CC note, and positions Series CC in relation to Series D (Paper 3's internal derivation notes) and Series T (the trilogy-wide concept-disambiguation notes). Together, these three series provide the complete prior-art framework for the trilogy.

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1. The Prior-Art Problem That Series CC Addresses

Series A and Series B establish Paper 1's and Paper 2's respective commitments as public prior art. Series D establishes Paper 3's commitments as derivations from its own six claims, covering Paper 3 from within—foundational sub-commitments, operational variants, anti-patterns, composition pairs, operational tests, and boundary cases. Series D is internal coverage: it answers the question of what

Paper 3 requires and why those requirements hold at inter-Self scope.

Series CC addresses a different vulnerability. An adversary seeking to claim novelty over some component of Paper 3's architecture does not need to dispute Paper 3 in isolation. The adversary's strongest move is to argue that Paper 3's inter-Self commitments are *categorically new*—that the shared substrate, Full Aspect Integration, three-tier conflict handling, the four-locus evolution-feed mechanism, configuration as substrate content, and population-scale collective evolution are conceptually disconnected from Papers 1 and 2 and therefore constitute independent inventions available for protection. If that argument lands, the adversary gets a gap in the prior-art chain even though the relevant commitments have been defended across all three papers.

Series CC forecloses this argument by establishing the inheritance chain explicitly and in advance. Each Series CC note takes one Paper 3 commitment and traces its lineage to the specific Paper 1 or Paper 2 commitment from which it extends. The chain is not asserted vaguely—it is demonstrated element by element, using a consistent structure that separates what each Paper 3 commitment preserved from its source from what it genuinely extended. An adversarial claim of novelty that fails to engage the inheritance chain fails to engage the prior art. Series CC makes the chain public, dated, and specific.

2. The Scope Extension Structure Underlying Series CC

The trilogy's three papers share a single architectural spine: substrate-mediated coordination under human governance, with the AI serving as substrate mediator rather than as a coordination oracle. Each paper extends this spine to a larger scope.

Paper 1 establishes the spine at cell scope. The substrate is the coordination medium within a single cell. The six Paper 1 commitments—substrate-based hybrid composition with a governance boundary, conflict preservation at the substrate level, human governance over content and rules, AI as substrate mediator, tool-agnosticism at the substrate layer, and linear-cost storage and composition—hold within the cell perimeter.

Paper 2 extends the spine to Self scope. A Self is the integration architecture unifying aspects and cells under a governing structure. Paper 2's instinct/reasoning separation places the LLM as the instinct layer and the CKS substrate as the reasoning layer within the Self's home perimeter. The six Paper 1 commitments carry forward into Paper 2's scope; Paper 2 adds the aspect as coordination unit, the DNA-layer/action-layer distinction, lifecycle and evolution operations, and the three-level cell/aspect/Self hierarchy.

Paper 3 extends the spine one further step, to the inter-Self perimeter. The shared substrate is the coordination medium between two or more CKS-governed Selves. All six Paper 1 commitments hold within the shared substrate's scope. Paper 2's aspect remains the unit of contribution to the shared substrate, and the instinct/reasoning separation continues to bound what content exchanges through the shared substrate (DNA-layer and action-layer content exchange; LLM weights and instinct-layer content do not). Paper 3 adds Full Aspect Integration as the canonical operation over the shared substrate, three-tier conflict handling, the four-locus evolution-feed mechanism, configuration as substrate content with recursive applicability, and population-scale collective evolution.

This layered structure is the foundation of Series CC's prior-art argument. Paper 3 does not depart from the spine; it extends it. The shared substrate is the Paper 1 substrate at inter-Self scope. Full Aspect Integration is the Paper 2 combination primitive at inter-Self scope. Three-tier conflict handling is Paper 1's conflict-preservation commitment extended to the inter-Self perimeter with explicit resolution and escalation tiers added. Every Paper 3 commitment has a specific source commitment in Paper 1 or Paper 2; Series CC names and formalizes each such source.

3. Series CC Structure: Two Subseries

Series CC comprises approximately forty notes organized as two subseries.

CC.1 — Paper 3 ↔ Paper 1 inheritance (~20 notes, #667–#686). CC.1 notes formalize the inheritance edge from each Paper 1 commitment to the corresponding Paper 3 extension. The expected inheritance edges include: Paper 3's shared substrate inheriting the substrate-as-coordination-medium commitment from Paper 1; Paper 3's conflict preservation within the shared substrate inheriting Paper 1's conflict-preservation commitment; Paper 3's human authority over the shared substrate's governance configuration inheriting Paper 1's human-governed commitment; Paper 3's AI-as-substrate-mediator role at the inter-Self perimeter inheriting Paper 1's AI-as-substrate-mediator commitment; Paper 3's tool-agnosticism at the shared substrate layer inheriting Paper 1's tool-agnosticism commitment; Paper 3's linear-cost composition at inter-Self scope inheriting Paper 1's linear-cost commitment; Paper 3's path retraceability across perimeters inheriting Paper 1's path-retraceability commitment; and the substrate-mediator ladder rungs 1 and 2 inheriting from Paper 1's cell-scope architecture as the foundation for the ladder's bottom rungs.

CC.2 — Paper 3 ↔ Paper 2 inheritance (~20 notes, #687–#705). CC.2 notes formalize the inheritance edge from each Paper 2 commitment to the corresponding Paper 3 extension. The expected inheritance edges include: Paper 3's aspect-as-exchange-unit inheriting from Paper 2's aspect-as-coordination-unit; Paper 3's FAI three pattern variants (full merge, governance-configured partial merge, lineage-preserved) inheriting from Paper 2's mating three pattern variants; Paper 3's instinct-takes-no-FAI-input commitment inheriting from Paper 2's instinct/reasoning separation; Paper 3's DNA-layer exchange through the shared substrate inheriting from Paper 2's DNA-layer structure; Paper 3's action-layer exchange inheriting from Paper 2's action-layer structure; Paper 3's evolution-feed routing to DNA evolution inheriting from Paper 2's DNA evolution mechanism; Paper 3's evolution-feed routing to action-feedback inheriting from Paper 2's action-feedback evolution mechanism; Paper 3's home-perimeter governance commitment inheriting from Paper 2's enterprise-Self governance architecture; and the substrate-mediator ladder rungs 3 and 4 inheriting from Paper 2's Self-scope architecture.

4. The Five-Element Structure of Each CC Note

Every CC.1 and CC.2 note follows a consistent five-element structure. The consistency is not stylistic convention—it is the mechanism that makes each inheritance claim precise enough to be useful for prior-art purposes.

Element 1: Cross-paper inheritance identification. The note opens by naming the source commitment (which Paper 1 or Paper 2 commitment is the source) and the target commitment (which Paper 3 commitment is the extension). This element establishes the specific inheritance edge being formalized.

Element 2: The source commitment at its original scope. The note states the Paper 1 or Paper 2 commitment in its original form, at its original scope. This element prevents vague inheritance arguments: the claim is not that Paper 3 generally derives from Papers 1 and 2 but that a specific commitment, at a specific scope, is the source.

Element 3: The extension to inter-Self scope. The note states how the Paper 3 commitment extends the source commitment to the inter-Self perimeter. This element names the scope change explicitly. What changed is the perimeter across which the commitment applies; the commitment itself is preserved.

Element 4: What is preserved versus what is extended. The note explicitly separates the preserved invariant (the commitment's structural content, which carries unchanged from the source paper) from the extension (the new scope, mechanism detail, or cardinality that Paper 3 adds). This separation is the analytical core of each CC note. Preserved invariants are the parts an adversary cannot claim as novel; extended elements are the parts Paper 3 adds—but added as scope applications of already-published commitments, not as conceptually new departures.

Element 5: Prior-art inheritance claim. The note closes with an explicit prior-art inheritance claim: the Paper 3 commitment is a derivation of the named source commitment; any party wishing to claim novelty over the Paper 3 commitment must first address the prior-art chain established by the source paper and this note.

This five-element structure means that every CC note contributes a discrete, verifiable prior-art claim. Reading all CC notes for a given Paper 3 commitment (which may span both a CC.1 and a CC.2 note, since some Paper 3 commitments inherit from both prior papers) gives the complete inheritance picture for that commitment.

5. Relationship to Series D

Series D covers Paper 3 from within. Its six phases—Phase D0 claim-level anchor notes, Phase D1 foundational sub-commitments, Phase D2 operational variants, Phase D3 anti-patterns, Phase D4 composition pairs, Phase D5 operational tests, and Phase D6 boundary cases—answer the question: given Paper 3's six claims, what does Paper 3 require, and why do those requirements hold at inter-Self scope?

Series CC covers Paper 3 from the outside. Its notes answer the complementary question: where did those requirements come from?

The two series together provide a complete prior-art foundation. A party reviewing only Series D can confirm that Paper 3's commitments are internally coherent and operationally sound at inter-Self scope. A party reviewing only Series CC can confirm that Paper 3's commitments are not novel departures from the prior trilogy papers but scope extensions of already-published commitments. A party reviewing both can confirm both internal coherence and external lineage—the complete picture.

The partitioning rule between Series D and Series CC is straightforward. A note belongs in Series D if its primary purpose is to defend a Paper 3 commitment against failure modes, anti-patterns, or boundary cases at inter-Self scope. A note belongs in Series CC if its primary purpose is to establish the inheritance edge from a Paper 1 or Paper 2 commitment to a Paper 3 commitment. Some composition pairs in Series D Phase D4 overlap conceptually with Series CC edges—when a composition pair involves a cross-paper pairing, the composition-pair defense goes in D4 and the inheritance-edge formalization goes in CC.

6. Relationship to Series T

Series T covers trilogy-wide concept disambiguation. Its notes address cases where the same concept appears in more than one of the three trilogy papers but is used at different scopes or with slightly different structural roles—cases where a reader might mistake a scope-specific usage for a conflict between papers.

The distinction between Series CC and Series T is the following. Series CC notes establish that Paper 3 extends a Paper 1 or Paper 2 commitment—they argue for continuity and derivation. Series T notes establish that the same concept used across papers is consistently defined but differently scoped—they argue against confusion or conflation. A Series CC note says: "Paper 3's three-tier conflict handling derives from Paper 1's conflict-preservation commitment." A Series T note on the same concept says: "Conflict preservation appears in Papers 1, 2, and 3, and in each case the structural commitment is the same; what differs is the perimeter scope at which the commitment operates."

The two series are complementary. CC establishes the lineage; T prevents conceptual confusion that could obscure the lineage. Together they support the complete trilogy prior-art claim: that the three papers are a coherent architectural lineage, not three independent and potentially competing descriptions of related architectures.

7. Three-Paper Attribution for Series CC Notes

Every CC.1 and CC.2 note carries three-paper attribution, citing all three trilogy papers. The attribution formula for Series CC is:

> *"This work derives from and formalizes commitments across the CKS theory trilogy: 'The Instinct/Reasoning Separation Outside the Model' (Li, April 2026), 'Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems' (Li, April 2026), and 'Inter-Self Coordination via Shared Substrate / Full Aspect Integration' (Li, April 2026)."*

This three-paper attribution reflects the structure of CC notes. A CC.1 note (P3 ↔ P1) draws directly from Papers 1 and 3, with Paper 2 present as context for the scope ladder and the inter-Self perimeter framing. A CC.2 note (P3 ↔ P2) draws directly from Papers 2 and 3, with Paper 1 present as the foundational layer whose commitments Papers 2 and 3 both carry. No CC note draws from only one or two trilogy papers in isolation; the inheritance chain is trilogy-wide even when a specific edge connects only two of the three papers.

8. Series CC in the Overall Derivation Note Structure

The full derivation note series addresses the CKS theory trilogy across six series. Series A (197 notes) and Series B (220 notes) cover Papers 1 and 2 internally. Phases A0 and B0 (6 notes each) anchor the six paper-level claims of Papers 1 and 2. Series C (30 notes) covers the Paper 1 ↔ Paper 2 cross-derivation. Series D (~200 notes) covers Paper 3 internally. Series CC (~40 notes, this series) covers Paper 3's inheritance from Papers 1 and 2. Series T (~50 notes) covers trilogy-wide concept disambiguation.

Within this structure, Series CC occupies a specific analytical layer. It is not the first place a reader encounters Paper 3's commitments—that is Series D Phase D0, which anchors Paper 3's six claims at paper-claim level. It is not the place to find Paper 3's operational tests or anti-pattern formalizations—those are in Series D Phases D5 and D3 respectively. Series CC's layer is the inheritance layer: the explicit, structured formalization of how the trilogy hangs together as an architectural lineage rather than three independent documents.

This note, CC.00, marks the beginning of Series CC at note #666 in the overall derivation note sequence. The CC.1 subseries begins at #667 and covers Paper 3 ↔ Paper 1 inheritance edges. The CC.2 subseries begins at approximately #687 and covers Paper 3 ↔ Paper 2 inheritance edges. Series T begins at approximately #706.

Source Papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems.* April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance.* April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *Inter-Self Coordination via Shared Substrate / Full Aspect Integration.* April 2026. ORCID: 0009-0004-8065-3235.

How to Cite This Note

Li, W. (2026). *Series CC Overview and Cross-Paper Inheritance Framework.* May 15, 2026. ORCID: 0009-0004-8065-3235.